

MATHEMATICAL LOGIC. — *Reversible Turing machines. Recursive insolubility in $n \in \mathbf{N}$ of the equation $u = \theta^n u$, where θ is an “isomorphism of codes.”* Note (*) by Mr. **Yves Lecerf**, presented by Mr. André Lichnerowicz.

We define “reversible Turing machines” and “isomorphisms of codes θ .” Their properties make it possible to prove that the equation in $n \in \mathbf{N}$, $u = \theta^n u$ is recursively unsolvable. A second note will apply this to the demonstration of a conjecture of Schützenberger relating the Post correspondence problem to the problem of diagonalization of homomorphisms of free monoids.

1. ISOMORPHISMS OF CODES, EPIMORPHISMS OF CODES. — *a. A conjecture of Schützenberger.* — Given two nontrivial free monoids $A^\dagger$ and $S^\dagger$, and given two homomorphisms φ and ψ of $A^\dagger$ into $S^\dagger$, consider the problem of the search for non-trivial solutions $x \in A^\dagger$ for the equation of diagonalization $\varphi x = \psi x$. A result of Post ⁽⁶⁾ is that this equation is recursively unsolvable in the case of φ and ψ being arbitrary homomorphisms. It is also so when one restricts φ to be a monomorphism; indeed, Chomsky and Schützenberger remarked ⁽¹⁾ that this case can be reduced to Post’s Tag-problem ⁽⁵⁾, itself recursively unsolvable according to a result of Minsky ⁽³⁾. Schützenberger conjectured that the equation $\varphi x = \psi x$ remains still recursively unsolvable when φ and ψ are both monomorphisms.

b. Isomorphisms of codes. — Instead of $\varphi x = \psi x$, it is equivalent to consider the equation $w = \theta w$, where $\theta = \psi\varphi^{-1}$ (this is shorthand notation for saying that θ is a bijection of $\varphi A^\dagger$ into $\psi A^\dagger$ defined by $\theta w = \psi x$ for $w = \varphi x$). For convenience, we will call the applications such as θ “isomorphisms of codes.” The term recalls that $\theta(w_1 w_2) = \theta w_1 \theta w_2$; and also that, $A = \{a_i\}_{i \in I}$ designating the alphabet (generators) of $A^\dagger$, $\{\varphi a_i\}_{i \in I}$ and $\{\psi a_i\}_{i \in I}$ are called “codes” on $S^\dagger$, because, for an arbitrary y in $S^\dagger$, there exists a set of indices $\{i_1, i_2, \dots, i_p\}$ such that $y = \varphi a_{i_1} \varphi a_{i_2} \dots \varphi a_{i_p}$, and the same for ψ . In fact, it is especially the study of isomorphisms of codes to which will be devoted the present Note and the following one.

c. Definitions of particular “isomorphisms of codes” using relation elements. — With e_A and e_S designating the identity elements respectively of $A^\dagger$ and $S^\dagger$, it goes without saying implicitly for every θ that one has $e_S = \theta e_A$, with $\varphi e_A = \psi e_A = e_S$ (whenceforth a trivial solution for $w = \theta w$ and for $w = \theta^n w$, with $n \in \mathbf{N}$). This being the case, each particular isomorphism of codes could be defined by a set of relation elements of the type $\{m_{i,\varphi} \rightarrow m_{i,\psi}\}_{i \in I}$, provided that $\{m_{i,\varphi}\}_{i \in I}$ and $\{m_{i,\psi}\}_{i \in I}$ are “codes” and that the correspondence is bijective. Indeed, $A^\dagger$ is implicitly defined by I , and $S^\dagger$ by the symbols used to note the $m_{i,\varphi}$ and $m_{i,\psi}$; and one can interpret the relations like correspondences $\{\varphi a_i \rightarrow \psi a_i\}_{i \in I}$.

d. *Checking whether a given set of words is a code.* — Further, the following property will be often called upon: If C and K_r designate respectively a code and a right prefix-code on $S^\dagger$, and if α is a symbol (generator of $S^\dagger$) not appearing in C nor in K_r , then, the set $C \cup \alpha K_r$ is a code. In the same way, replacing K_r by a left prefix-code K_ℓ , the set $C \cup K_\ell \alpha$ is a code. Let us recall that any right prefix-code K_r is by definition ⁽⁴⁾ such that, if $m_i, m_j \in K_r$ and if, with $y \in S^\dagger$, one has $m_i = m_j y$, then $y = e_S$ (while for the left prefix-codes, it is $m_i = y m_j$ which imposes $y = e_S$).

e. *Epimorphisms of codes.* — One speaks about “epimorphisms of codes” τ in the case of relations $\{m_{i,\varphi} \rightarrow m_{i,\psi}\}_{i \in I}$, where $\{m_{i,\varphi}\}_{i \in I}$ is a complete code C_φ , but where $\{m_{i,\psi}\}_{i \in I}$ is only constrained not to contain words other than those of a code C_ψ .

2. REVERSIBLE TURING MACHINES. — Let MT be a Turing machine of which $\{\varepsilon_p\}_{p \in P}$ and $\{\sigma_q\}_{q \in Q}$ are the sets of states and symbols, and $\{\delta_r\}_{r \in R}$ are tape displacements, which can be ± 1 or 0. One can define MT by a set of quintuples

$$\chi_{MT} = \{\varepsilon_{p_1(i)}; \sigma_{q_1(i)}; \varepsilon_{p_2(i)}; \sigma_{q_2(i)}; \delta_{r(i)}\}_{i \in I},$$

where the indices p_1, p_2, q_1, q_2, r are functions of index i . With each of the quintuples, let us decide to associate an “inverse image quintuple” $(\varepsilon_{p_2(i)}^*; \sigma_{q_2(i)}; \varepsilon_{p_1(i)}^*; \sigma_{q_1(i)}; -\delta_{r(i)})$. The set of those will generally not constitute a Turing machine; but when it does, we will say that MT is “reversible,” and call the new machine the inverse image MT^* of MT. The ε_p^* will be known as the images of ε_p . The substitution of ε_p^* for ε_p in an instantaneous configuration U_k will be known as transformation of U_k to its image configuration U_k^* . The continuations of configurations of MT^* are images of those of MT, but MT^* traverses them in the opposite order. Now let us consider the machine $R(MT)$, whose set of quintuples is

$$\chi_{R(MT)} = \chi_{MT} \cup \chi_{MT}^* \cup \{(\varepsilon_p; \sigma_q)_{\text{halt}}; \varepsilon_p^*; \sigma_q; 0\},$$

where $(\varepsilon_p; \sigma_q)_{\text{halt}}$ designates any state-symbol pair for which MT halts. If one starts MT and $R(MT)$ from the same instantaneous configuration U_0 , they pass through the same configurations as long as MT does not halt (thus possibly indefinitely). When MT halts, $R(MT)$ continues, traversing in the opposite order the image configurations of the traversed configurations, and passes by the image of the initial configuration. $R(MT)$ will be known as the coupling of MT with its reverse image.

3. REPRESENTATION OF TURING MACHINES BY EPIMORPHISMS (OR ISOMORPHISMS) OF CODES. — Let us be given an arbitrary MT. With each quintuple having movement $+1$, that is to say for example $(\varepsilon_g, \sigma_h, \varepsilon_j, \sigma_k, 1)$, we associate three relation elements, namely: $\{\alpha_g \sigma_h \rightarrow \sigma_k \alpha_j; \omega_g \sigma_h \rightarrow \sigma_k \alpha_j; \sigma_h \beta_g \rightarrow \sigma_k \alpha_j\}$. With $(\varepsilon_g, \sigma_h, \varepsilon_j, \sigma_k, 0)$, we associate: $\{\alpha_g \sigma_h \rightarrow \omega_j \sigma_k; \omega_g \sigma_h \rightarrow \omega_j \sigma_k; \sigma_h \beta_g \rightarrow \omega_j \sigma_k\}$. With $(\varepsilon_g, \sigma_h, \varepsilon_j, \sigma_k, -1)$, we associate $\{\alpha_g \sigma_h \rightarrow \beta_j \sigma_k; \omega_g \sigma_h \rightarrow \beta_j \sigma_k; \sigma_h \beta_g \rightarrow \beta_j \sigma_k\}$. Finally, with any symbol σ_q of MT, we associate $\sigma_q \rightarrow \sigma_q$. One can check, by the process given in paragraph 1d, that the set of these relations defines an epimorphism of codes. With $\tau_{\max}$ being this set, $\tau_{\max}$ is a representation of MT, because it defines its alphabet and quintuples. We can, in addition, find, for the instantaneous configurations of MT, notations such that for any pair of successive configurations u_i, u_{i+1}

we have $u_{i+1} = \tau_{\max} u_i$. For that, a configuration will be composed of a succession of symbols σ (the string on the tape) into which one will intercalate one of the letters α , ω or β , with an index p equal to that of the state ε_p of the machine, and indicating, not only the position π_1 of the next symbol to read, but also the position π_2 of the symbol previously written (with a particular convention for the initial configuration). An α_p signifies that π_1 is the first symbol to its right, π_2 the first on its left. A β_p , vice-versa. An ω_p means that π_1 and π_2 are both the first symbol to the right of the ω_p . We have then achieved that $u_{i+1} = \tau_{\max} u_i$. So certain states ε_p can appear under only two or one of the forms α_p , ω_p , β_p , and $\tau_{\min}$ is obtained by removing from $\tau_{\max}$ all the relation elements containing the forms which never appear, so $\tau_{\min}$ is still such that $u_{i+1} = \tau_{\min} u_i$. If $\tau_{\min}$ is an isomorphism of codes, MT is reversible.

4. SIMULATION OF ARBITRARY MT ON REVERSIBLE MT'. APPLICATION TO ISOMORPHISMS OF CODES. — *a. Properties.* — One can simulate an arbitrary Turing machine MT (with configurations v_i) on a reversible Turing machine MT_ρ (with configurations $u_{i,j}$) so that: (1) when MT passes from v_i to v_{i+1} , MT_ρ passes from $u_{i,0}$ to $u_{i+1,0}$ via the intermediary of a finite number of configurations $u_{i,1}; u_{i,2}; \dots$; (2) we pass from one v_i to the next via an epimorphism of codes τ , and from one $u_{i,j}$ to the next via an isomorphism of codes θ ; (3) if the initial configurations are v_0 for MT and $u_{0,0}$ for MT_ρ , with $u_{0,0} = \lambda v_0 \mu \nu$, then for any i , one has $u_{i,0} = \lambda v_i \mu w_i \nu$, where w_i is a string, and where λ, μ, ν are three symbols which appear neither in v_i nor in w_i , so that knowing $u_{i,0}$ gives v_i and w_i ; (4) there are symbols r_k of which each one represents a relation element of τ other than that of identity; a blank symbol b ; and for any i we have $w_i = b^2 r_{k_1} r_{k_2} \dots r_{k_i} b$, where r_{k_p} is the relation invoked by $v_p = \tau v_{p-1}$. Thus, w_i represents the history of the computation of MT until time i ; (5) MT_ρ halts on the $u_{i,0}$ corresponding to the halting of MT, and them only; (6) the machine $R(MT_\rho)$, coupling MT_ρ with its reverse image, starting from $u_{0,0} = \lambda v_0 \mu \nu$ passes through the image configuration $\lambda v_0^* \mu \nu$ if and only if MT, starting from v_0 , halts; (7) there exists for $R(MT_\rho)$ certain instantaneous configurations $u_{s,t}$ such that, when started at $\lambda v_0 \mu \nu$, $R(MT_\rho)$ cannot reach those configurations other than by passing through $\lambda v_0^* \mu \nu$ (i.e., if MT, starting from v_0 , halts). One can thus arrange that the return of $R(MT_\rho)$ to $\lambda v_0 \mu \nu$ (or the passage through $u_{s,t}$ framed by $\lambda' \nu'$ instead of $\lambda \nu$) is conditional on the halting of MT.

Proof. — It is shown how to proceed from τ , presumed to be given by a set of relation elements $\{I_{k,\tau}\}_{k \in K_\tau}$ to the set of relation elements $\{I_{j,\theta}\}_{j \in J_\theta}$ defining θ and MT_ρ . We delimit the principles of this construction, by showing how to simulate an $I_{k_i,\tau}$ of the form $\alpha_p \sigma_q \rightarrow \sigma_f \alpha_g$. With this, we associate: an instruction $\alpha_p \sigma_q \rightarrow \sigma_{f,g,\alpha} \varepsilon_{\alpha,\alpha,p,q,f,g,\sigma}$, where the symbol $\sigma_{f,g,\alpha}$ marks the place where one must modify v_i , and the nature of the modification; instructions allowing control to be lead to the left from ν through a state $\varepsilon_{\alpha\alpha p q f g \nu}$; an instruction $b \varepsilon_{\alpha\alpha p q f g \nu} \rightarrow \varepsilon_s r_{k_i}$, where r_{k_i} represents $I_{k_i,\tau}$, supplementing w_i ; working instructions moving ν and possibly also λ, μ and the entire w_i , to restore the necessary blanks in $u_{i,0}$ and then defer control in $\sigma_{f,g,\alpha}$ with a state ε_σ ; an instruction $\sigma_{f,g,\alpha} \varepsilon_\sigma \rightarrow \sigma_f \alpha_g$ which supplements v_i in $u_{i,0}$.

b. THEOREM 1. — *The halting problem for a general reversible Turing machine is undecidable. Similarly for the problem of returning to the initial configuration, and*

that of the passage through a given configuration other than the initial configuration.

c. THEOREM 2. — The equation $w = \theta^n w$, where θ is an isomorphism of codes, with $n \in \mathbf{N}$ is recursively unsolvable in n given arbitrary w, θ . The equation $w_1 = \theta^n w_2$, with $w_1 \neq w_2$, is also recursively unsolvable in n .

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(¹) N. CHOMSKY et M. P. SCHÜTZENBERGER, *Computer Programming and Formal Systems*, Hirschberg and Braffort, North-Holland Publ. Co., Amsterdam, 1963, p. 118–161.

(²) H. WANG, *Mathematische Annalen*, **152**, 1963, p. 65–74.

(³) M. MINSKY, *Annals of Math.*, **74-3**, 1961.

(⁴) M. P. SCHÜTZENBERGER, *I. R. E. Trans. Inf. Theory*, IT-2, 1956, p. 47–60.

(⁵) E. POST, *Amer. J. Math.*, **65**, 1943, p. 196–215.

(⁶) E. POST, *Bull. Amer. Math. Soc.*, **52**, 1946, p. 264–268.

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